

Artificial Intelligence

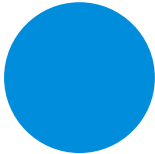


Course

By Dr. Fotsing Kuetché



Fall 2025



Why This Course ?



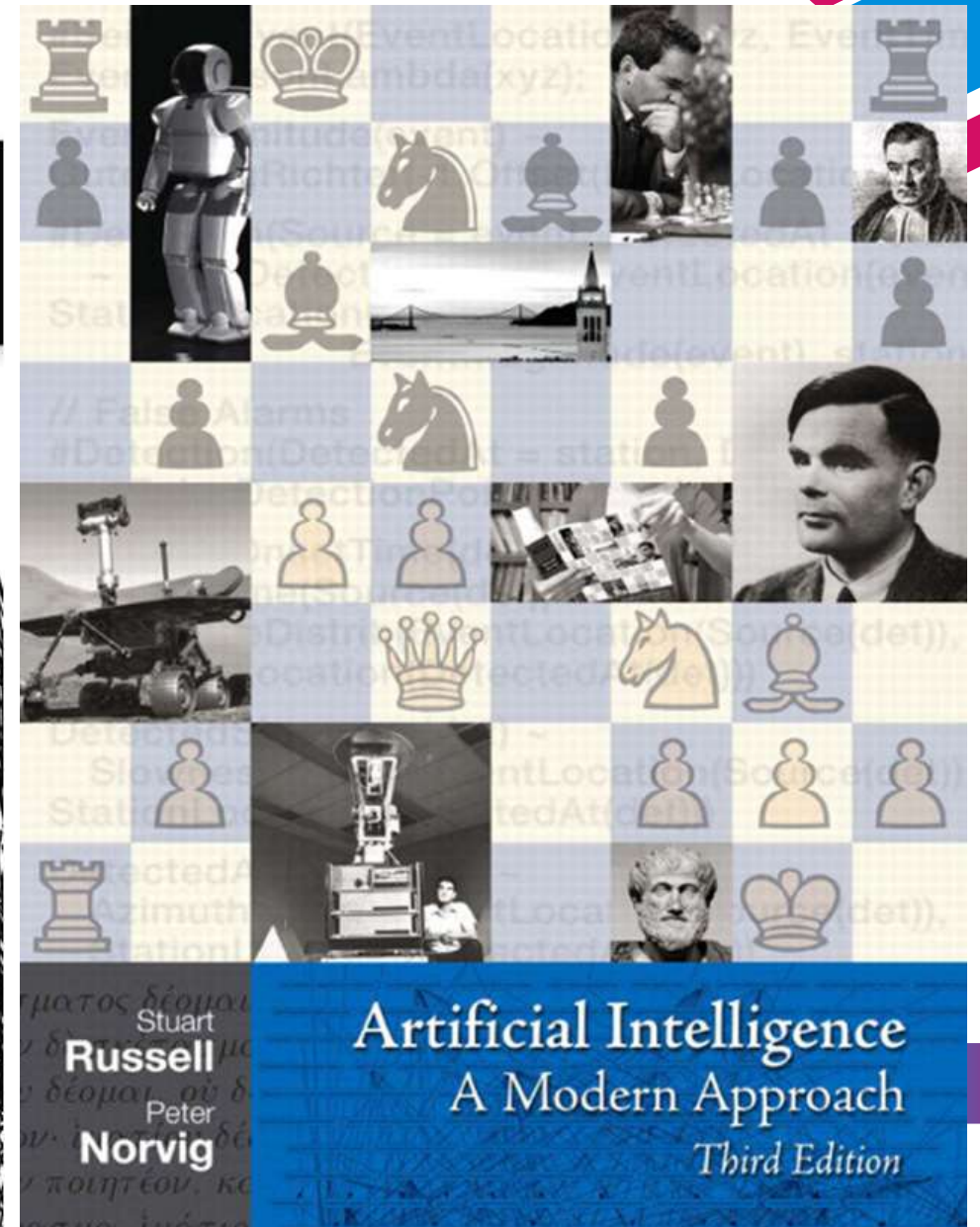
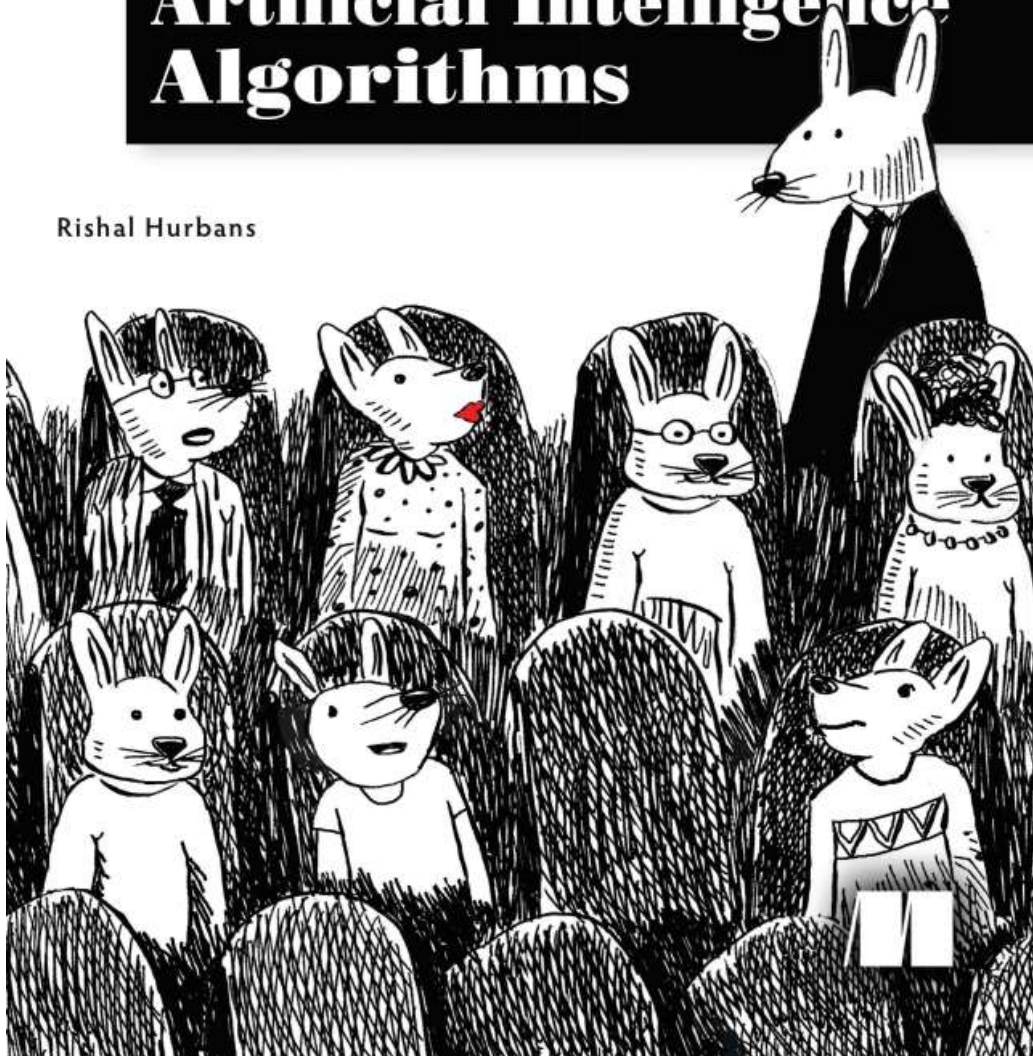
- Introduce the fundamental concepts and history of Artificial Intelligence.
- Explain the various problem-solving paradigms and intelligent search methods.
- Explore heuristic and metaheuristic algorithms for optimization problems.
- Present key concepts of evolutionary computation and swarm intelligence.
- Introduce machine learning principles, including supervised and unsupervised learning.
- Familiarize students with artificial neural networks and learning algorithms.
- Enable students to apply AI techniques to solve real-world problems.

Main Textbooks

grokking

Artificial Intelligence Algorithms

Rishal Hurbans



what is Artificial Intelligence?



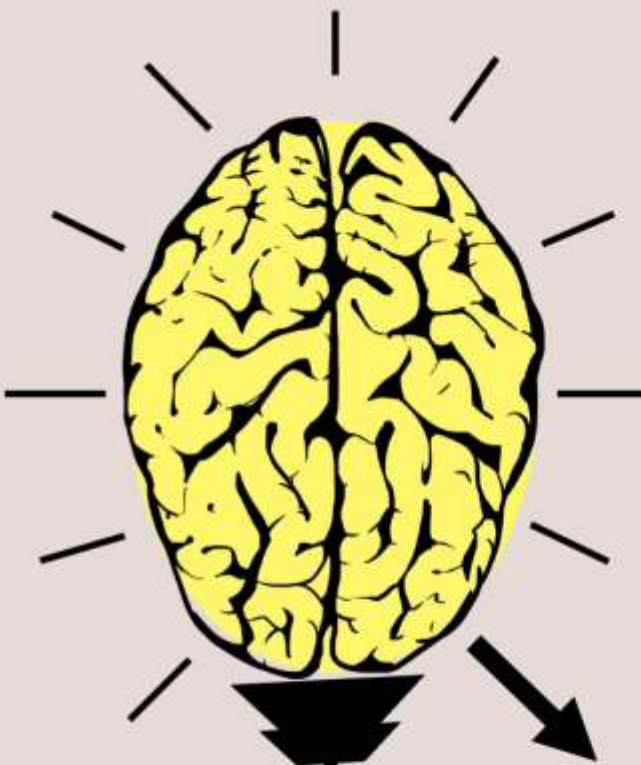
Artificial Intelligence has transformed from science fiction into an integral part of our daily lives. From voice assistants to recommendation systems, AI is reshaping how we interact with technology and solve complex problems.



What's AI ?

Understanding the fundamentals





Intelligence:

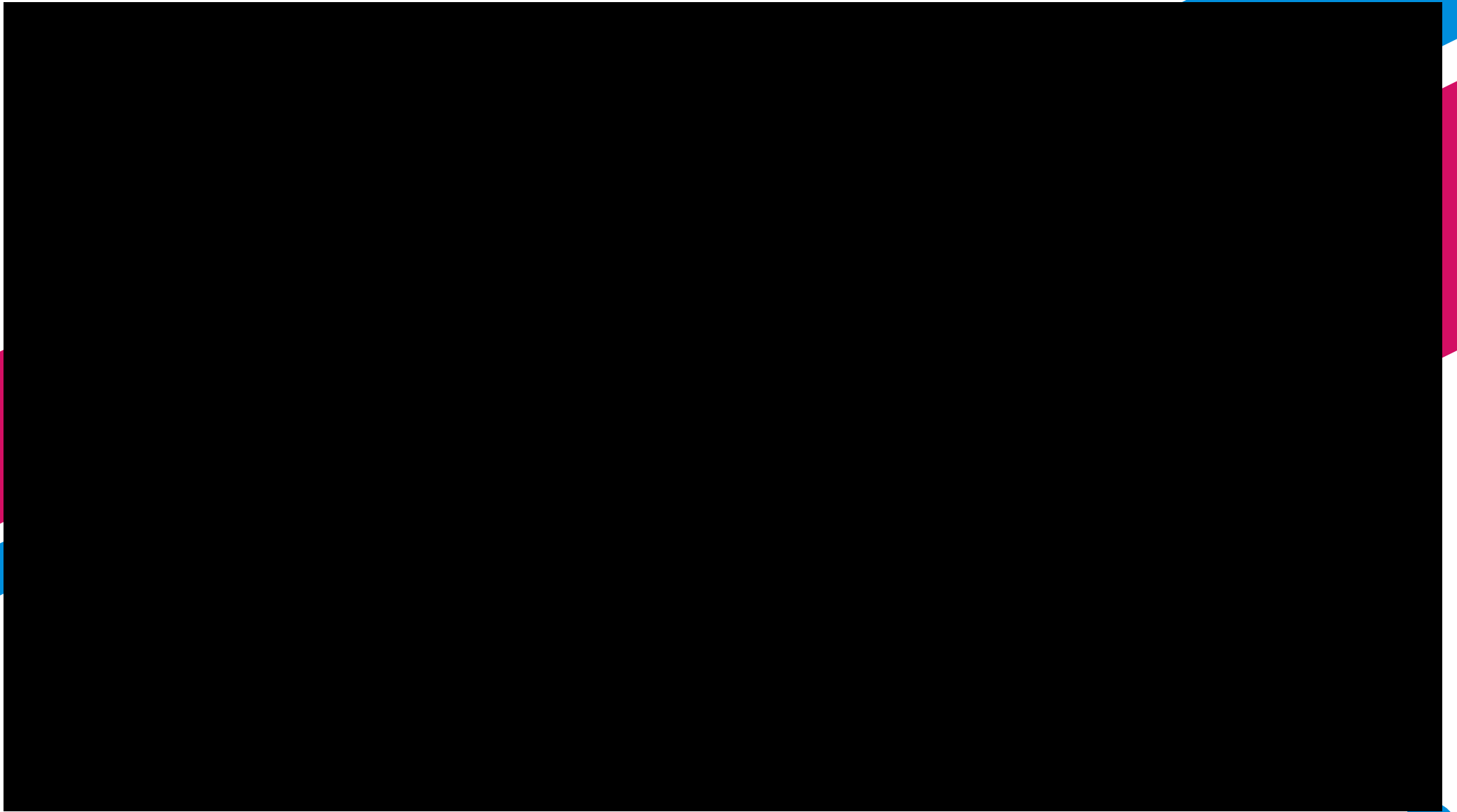
5-Minute Psych

What's Intelligence ?

What should we remember ?



1. Alfred Binet created the first standardized intelligence test to measure "mental age," helping to identify children needing extra educational support.
 2. The intelligence quotient (IQ) was introduced by William Stern, calculated as $(\text{mental age} / \text{chronological age}) \times 100$, where 100 is average.
 3. Charles Spearman proposed that intelligence comprises specific abilities (S factors) as well as a general intelligence factor (g factor) that underlies all cognitive tasks.
 4. We have verbal and non-verbal intelligence
 5. Intelligence has been categorized into crystallized intelligence (knowledge from past learning) and fluid intelligence (problem-solving and reasoning abilities in new situations), with crystallized intelligence rising with age while fluid tends to decline after early adulthood.
 6. Howard Gardner's theory of multiple intelligences expands the concept beyond traditional measures, including linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, interpersonal, naturalistic, and existential intelligences.
- 



What's Intelligence ?

What should we remember ?



1. Intelligence was once thought to be uniquely human, setting humans apart from all other species.
 2. Charles Darwin challenged this view by showing intelligence evolved through natural selection and is shared among many species.
 3. Early animal behaviorists believed animal actions were mere stimulus-response, but this view was overturned by detailed observations.
 4. Jane Goodall's work with chimpanzees demonstrated that tool-making is not exclusive to humans.
 5. Many animals, including whales, dolphins, crows, octopuses, and scrub jays, display advanced cognitive abilities like communication, tool use, problem-solving, and memory.
 6. In some cognitive tasks, certain animals even outperform humans, such as chimpanzees' superior short-term memory for numbers.
 7. Human intelligence is still special but not uniquely superior; the intelligence landscape is more diverse and complex than previously thought.
- 

What's Intelligence ?



Intelligence is a mystery—a concept that has no agreed-upon definition.

Philosophers, psychologists, scientists, and engineers all have different opinions about what it is and how it emerges



The Notion of Intelligence



Human Intelligence

The ability to learn from experience, adapt to new situations, understand complex concepts, and apply knowledge to manipulate one's environment.

- Problem-solving abilities
- Abstract reasoning
- Emotional understanding
- Creative thinking



Super Intelligence

Definition

An intellect that vastly surpasses the cognitive performance of humans in virtually all domains of interest, including scientific creativity, general wisdom, and social skills.

Potential Impact

Could revolutionise scientific discovery, solve global challenges, and fundamentally alter human civilisation—but also raises profound ethical and safety questions.

Timeline Debate

Experts disagree on when—or if—superintelligence will emerge, with predictions ranging from decades to centuries, making preparation crucial.



General Intelligence



Versatile Cognition

The capacity to understand, learn, and apply **knowledge** across diverse domains without being limited to specific tasks or narrow contexts.



Transfer Learning

The ability to take **knowledge** gained in one area and successfully apply it to solve problems in entirely different domains—something humans excel at naturally.

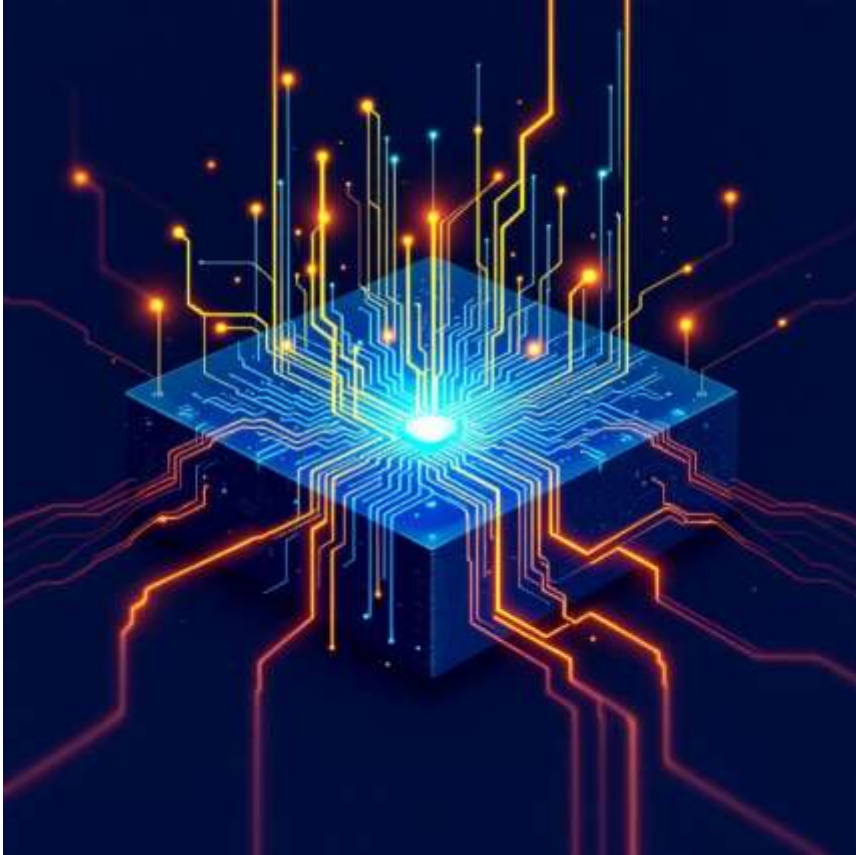


Adaptive Thinking

General intelligence enables flexible problem-solving, allowing agents to navigate unfamiliar situations and develop novel solutions without explicit programming.



What is Artificial Intelligence?



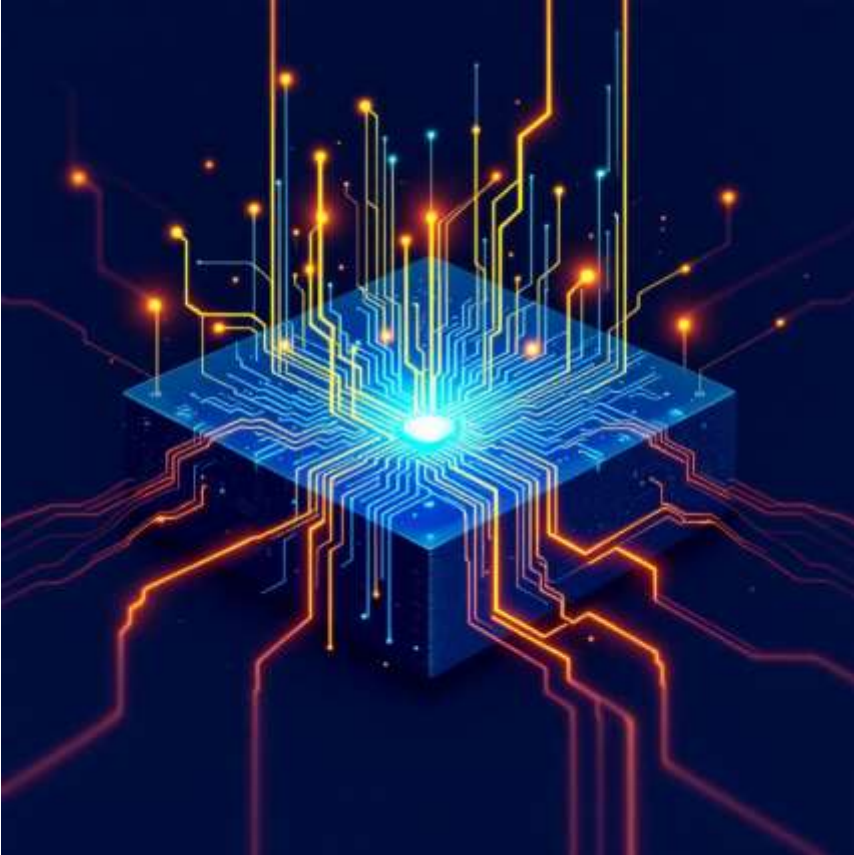
Artificial Intelligence represents the development of computer systems capable of performing tasks that typically **require human intelligence**.

simulate cognitive functions such as

- learning from experience,
- recognising patterns,
- making decisions
- understanding natural language.

AI involves creating algorithms and models that enable machines to process information, reason about it, and generate appropriate responses—mirroring the remarkable capabilities of the human mind.

What is Artificial Intelligence?



AI involves creating **algorithms and models** that enable machines to :

- process information,
- reason about it,
- and generate appropriate responses—

mirroring the remarkable capabilities **of the human mind**.

Key AI Capabilities

Learning

AI systems acquire knowledge through data analysis and experience, improving performance over time without explicit reprogramming.

Reasoning

Drawing logical conclusions from available information, enabling AI to make informed decisions and solve complex problems systematically.

Problem Solving

Identifying challenges, analysing constraints, and developing optimal solutions through algorithmic approaches and heuristic methods.

Perception

Interpreting sensory data from the environment, including visual recognition, speech understanding, and pattern detection in complex datasets.

Language Understanding

Comprehending and generating human language, enabling meaningful communication between humans and machines across multiple languages.



Types of Artificial Intelligence



Narrow AI

Specialised Intelligence

Systems designed to perform specific tasks exceptionally well but cannot transfer their expertise to other domains. Also known as Weak AI.

- Voice assistants (Siri, Alexa)
- Recommendation algorithms
- Image recognition systems
- Chess-playing programmes

Status: Currently deployed worldwide



General AI

Human-Level Intelligence

Hypothetical systems with cognitive abilities matching human intelligence across all domains, capable of understanding, learning, and applying knowledge flexibly.

- Context understanding
- Abstract reasoning
- Emotional intelligence
- Creative problem-solving

Status: Theoretical; not yet achieved

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The Birth of AI

1940s-1950s: Theoretical Foundations

- ✓ Alan Turing proposed the concept of machine intelligence and the famous Turing Test,

1950s-1960s: Early Programmes

Development of the Logic Theorist and General Problem Solver demonstrated that machines could engage in **symbolic reasoning** and prove mathematical theorems.

1

2

1956: The Dartmouth Conference

- John McCarthy coined the term "Artificial Intelligence" at this landmark gathering,
- Marvin Minsky and Herbert Simon launched AI as a formal academic discipline.

3



Key AI Milestones



1950: The Turing Test

Alan Turing proposed a test for machine intelligence, asking whether computers could exhibit behaviour indistinguishable from humans in conversation.



1990s-2000s: Machine Learning Emerges

Statistical approaches enabled computers to learn from data. IBM's Deep Blue defeated world chess champion Garry Kasparov in 1997, marking a watershed moment.



1970s-1980s: Expert Systems

Rule-based systems like MYCIN demonstrated AI's practical applications in medicine and industry, encoding human expertise into logical frameworks.



2010s: Deep Learning Revolution

Neural networks with multiple layers achieved breakthrough performance in image recognition, natural language processing, and game-playing, transforming AI capabilities.



The Modern AI Era

Deep Learning Dominance

Today's AI landscape is defined by deep neural networks that learn hierarchical representations from vast datasets, enabling unprecedented accuracy in perception and prediction tasks.

- Computer vision and facial recognition
- Natural language models (GPT, BERT)
- Autonomous vehicles
- Medical diagnosis assistance

AI Everywhere

Artificial Intelligence has become embedded in everyday applications, from personalised content recommendations to smart home devices, financial fraud detection, and climate modelling.

The future: AI continues evolving rapidly, with ongoing research in explainability, ethics, and achieving more generalised intelligence.



Problem Types in Artificial Intelligence

Well-Defined Problems

Clear objectives with specific rules and measurable outcomes. Examples include chess games, mathematical calculations, and route optimisation, where success criteria are unambiguous and verifiable.

Ill-Defined Problems

Ambiguous goals with multiple valid solutions and subjective evaluation criteria. Creative tasks, social interactions, and ethical dilemmas fall into this category, requiring nuanced interpretation and contextual understanding.

Ask to chatGPT to create an image of an ideal world



Problem Types in Artificial Intelligence

- 
1. Search problems: Find a path to a solution
 2. Optimization problems: Find a good solution
 3. Prediction and classification problems: Learn from patterns in data
 4. Clustering problems: Identify patterns in data
- 

AI Problem-Solving Paradigms

Artificial intelligence employs diverse methodologies to tackle complex challenges, each paradigm offering unique strengths suited to different problem domains.



Search Algorithms

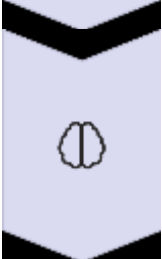
Systematic exploration of possible solutions through state spaces, using techniques like breadth-first, depth-first, and A* search to navigate problem landscapes efficiently.



Rule-Based Systems



Heuristics



Machine Learning



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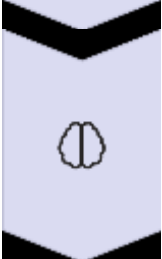


Rule-Based Systems

Logical frameworks using if-then statements to encode domain expertise, enabling systems to reason through problems using predefined knowledge structures.



Heuristics



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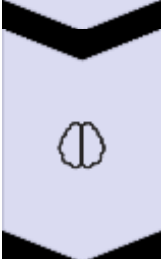


Rule-Based Systems



Heuristics

Practical shortcuts and rules of thumb that guide decision-making when exhaustive search is impractical, trading guaranteed optimality for computational efficiency.



Machine Learning



AI Problem-Solving Paradigms

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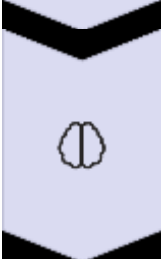
Search Algorithms



Rule-Based Systems



Heuristics



Machine Learning

Data-driven approaches that discover patterns and build models through experience, enabling systems to improve performance without explicit programming.



AI Subdomains Overview

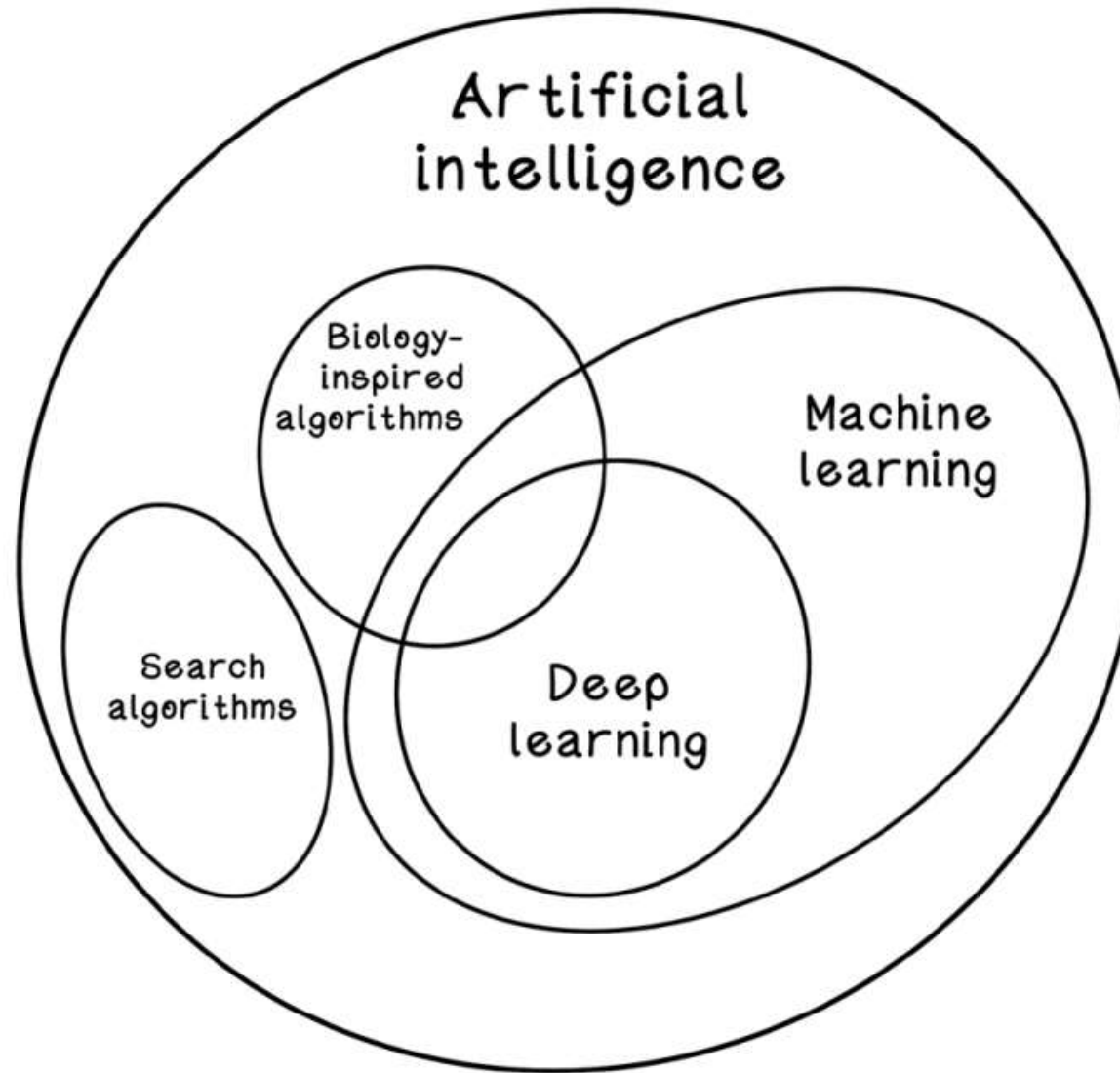
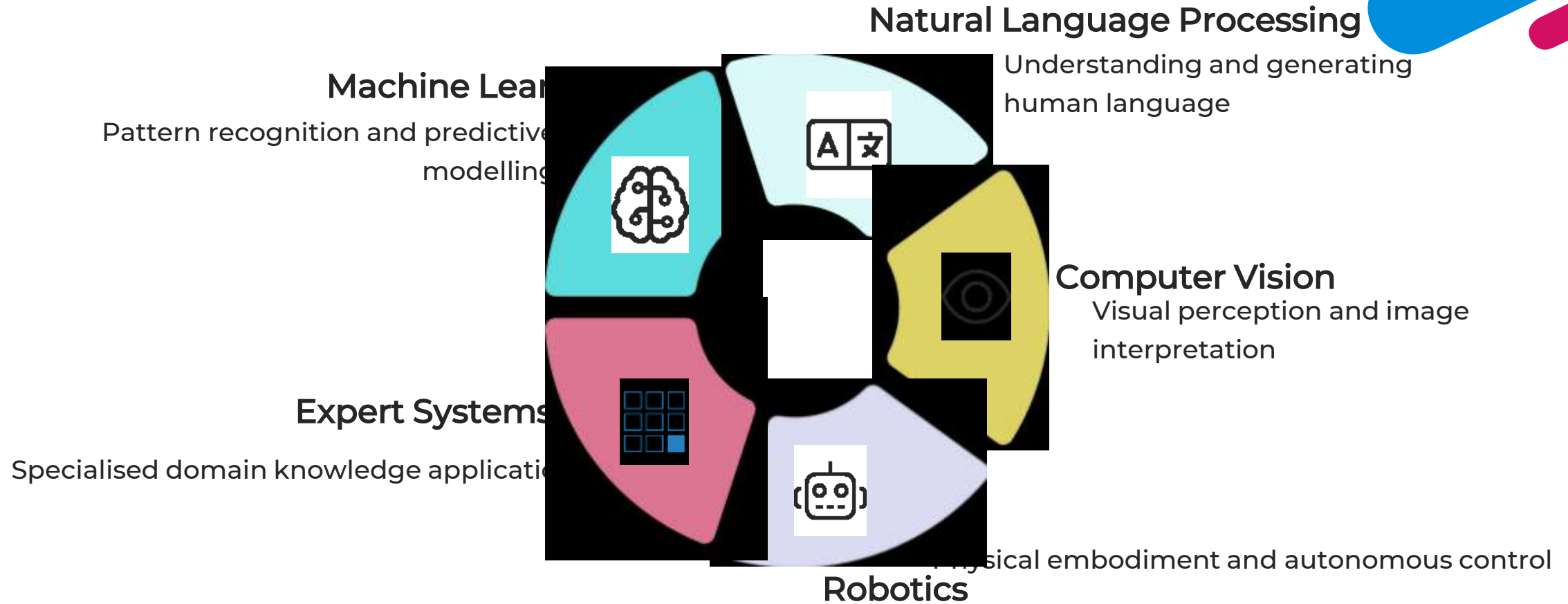


Figure 1.7 Categorization of concepts within AI

AI Subdomains Overview



These subdomains often intersect and complement one another, creating powerful hybrid systems that leverage multiple approaches to solve real-world challenges.

Machine Learning

Algorithms That Learn From Data

a paradigm shift in computing: rather than explicitly programming every rule, systems automatically discover patterns and relationships within data to make predictions and decisions.

Core Principles:

- Training on large datasets to identify underlying patterns
- Generalising from examples to handle new, unseen situations
- Continuous improvement through feedback and additional data

This approach powers everything from recommendation engines to autonomous vehicles, fundamentally transforming how computers solve complex problems.



Supervised Learning

Learning from labelled examples with known outcomes

Unsupervised Learning

Discovering hidden structures in unlabelled data

Reinforcement Learning

Learning through trial, error, and reward signals

Natural Language Processing

bridges the gap between human communication and computer understanding, enabling machines to comprehend, interpret, and generate human language with increasing sophistication.



Language Understanding

Parsing syntax, extracting meaning, and resolving ambiguity in written and spoken text through grammatical analysis and semantic interpretation.



Text Generation

Creating coherent, contextually appropriate written content ranging from simple responses to complex articles and creative writing.



Dialogue Systems

Engaging in natural conversations, understanding context, maintaining coherence across exchanges, and responding appropriately to user inputs.

Modern NLP systems leverage transformer architectures and large language models trained on vast text corpora, achieving near-human performance on many linguistic tasks.



Computer Vision

- ✓ empowers machines to interpret and understand visual information from the world,
- ✓ replicating and sometimes surpassing human visual perception capabilities.

01

Image Acquisition

Capturing visual data through cameras and sensors

02

Pre-processing

Enhancing image quality and extracting relevant features

03

Analysis & Recognition

Identifying objects, patterns, and relationships within images

04

Interpretation

Understanding context and deriving meaningful insights

Key Applications:

Facial recognition,
autonomous vehicle navigation,
medical image analysis, quality control in manufacturing, and augmented reality experiences.



Robotics



physical embodiment of artificial intelligence, where computational intelligence meets mechanical systems to create autonomous agents capable of interacting with the physical world.

Perception & Sensing

Gathering environmental data through cameras, lidar, touch sensors, and proprioceptive feedback to build spatial awareness.

Planning & Control

Determining optimal actions, generating movement trajectories, and executing precise motor commands whilst adapting to dynamic conditions.

Human-Robot Interaction

Safely collaborating with people through natural interfaces, gesture recognition, and adaptive behaviours that respond to human presence.



Expert Systems



Mimicking Human Expert Decision-Making

encode specialised knowledge from human experts into computational frameworks, enabling machines to reason through complex domain-specific problems with professional-level competence.

combine a knowledge base containing domain facts and rules with an inference engine that applies logical reasoning to reach conclusions and provide explanations for their recommendations.

Knowledge Acquisition

Extracting expertise from domain specialists

1

Inference

Applying logical reasoning to derive conclusions

3

Knowledge Representation

Encoding rules and facts in structured formats

2

Explanation

Justifying recommendations with transparent logic

4

Application of AI





Applications of Artificial Intelligence

Transforming Industries

transcended theoretical research - transformative force across virtually every sector of modern society, reshaping how we work, communicate, and solve problems.

\$15.7T

Projected Global Economic Impact

Estimated AI contribution to the global economy by 2030

97M

New Jobs Created

Employment opportunities emerging from AI technologies

70%

Businesses Adopting AI

Companies integrating AI into operations by 2025



From healthcare and finance to transportation and entertainment, AI applications are enhancing decision-making, automating routine tasks, and unlocking previously impossible innovations.

AI in Healthcare

Artificial intelligence is revolutionising healthcare delivery, improving diagnostic accuracy, personalising treatment approaches, and accelerating the development of life-saving therapies.



Advanced Diagnostics

AI-powered image analysis detects diseases like cancer, diabetic retinopathy, and cardiovascular conditions with accuracy matching or exceeding specialist physicians. Machine learning algorithms identify subtle patterns in medical imaging that human eyes might miss, enabling earlier intervention and improved outcomes.



Personalised Medicine

By analysing genetic profiles, lifestyle factors, and treatment responses, AI systems tailor therapeutic strategies to individual patients. This precision approach optimises medication selection, dosing, and treatment protocols whilst minimising adverse effects and maximising efficacy.



Accelerated Drug Discovery

AI dramatically reduces the time and cost of developing new pharmaceuticals by predicting molecular interactions, identifying promising drug candidates, and optimising clinical trial designs. What once took years can now occur in months, bringing life-saving treatments to patients faster.

AI in Finance

Artificial intelligence is revolutionising the financial sector, transforming how institutions detect fraud, execute trades, and assess creditworthiness. These technologies process vast amounts of data at unprecedented speeds, enabling more accurate predictions and efficient operations.



Fraud Detection

Machine learning algorithms analyse transaction patterns in real-time, identifying suspicious activities and preventing fraudulent transactions before they occur. These systems continuously learn from new data, adapting to evolving fraud tactics.



Algorithmic Trading

AI-powered trading systems execute high-frequency trades based on complex market analysis, processing millions of data points to identify profitable opportunities within milliseconds and optimise portfolio performance.



Credit Scoring

Advanced models assess creditworthiness by analysing diverse data sources beyond traditional metrics, enabling fairer lending decisions and expanding access to financial services for underserved populations.

AI in Retail and E-Commerce

The retail landscape has been transformed by artificial intelligence, creating personalised shopping experiences that anticipate customer needs and preferences. From recommendation engines to conversational interfaces, AI is reshaping how consumers discover and purchase products online.



Recommendation Systems

Sophisticated algorithms analyse browsing history, purchase patterns, and user behaviour to suggest products tailored to individual preferences. These systems increase conversion rates whilst enhancing customer satisfaction through personalised discovery.

- Collaborative filtering based on similar users
- Content-based recommendations matching preferences
- Hybrid approaches combining multiple techniques
- Real-time adaptation to changing behaviours

Intelligent Chatbots

Natural language processing enables conversational AI assistants that handle customer enquiries 24/7, resolving issues instantly and escalating complex queries to human agents when necessary.

Inventory Optimisation

Predictive analytics forecast demand patterns, helping retailers maintain optimal stock levels, reduce waste, and ensure product availability during peak periods.

AI in Transportation

Transportation is experiencing a profound transformation through artificial intelligence, promising safer roads, reduced congestion, and more efficient mobility solutions. From self-driving vehicles to intelligent traffic systems, AI is reshaping how we move through our cities and beyond.

Autonomous Vehicles

Self-driving cars use computer vision, sensor fusion, and deep learning to navigate complex environments, recognising pedestrians, traffic signals, and obstacles whilst making split-second driving decisions.

Smart Traffic Management

AI-powered systems analyse traffic flow in real-time, adjusting signal timings and suggesting alternative routes to reduce congestion, minimise emissions, and improve overall urban mobility.

Predictive Maintenance

Machine learning algorithms monitor vehicle health and infrastructure conditions, predicting maintenance needs before failures occur, reducing downtime and enhancing safety across transport networks.

AI in Natural Language Processing



Digital Assistants

Virtual assistants like Siri, Alexa, and Google Assistant understand spoken commands, answer queries, and perform tasks through sophisticated natural language understanding and generation capabilities.

Intelligent Chatbots

Conversational AI powers customer service platforms that comprehend context, sentiment, and intent, providing human-like interactions that resolve enquiries efficiently across multiple languages and channels.

Text Analysis

NLP systems extract insights from unstructured text, performing sentiment analysis, topic modelling, and information extraction to help organisations understand customer feedback and market trends.

Natural language processing bridges the gap between human communication and machine understanding, enabling computers to interpret, generate, and respond to text and speech in ways that feel increasingly natural and intuitive.

